PHYS7045/4045 Assignment 3

Due 30/10/2025 5 PM

Note: You may use Mathematica or other computer languages to help you solve these problems. If you do, you must show your code and describe your reasoning to get full marks.

Do not use generative AI to assist you in this assignment.

Please upload your scanned or typeset solutions to Blackboard. The upload should:

- *be a SINGLE pdf file (not multiple files).*
- *include your name in the file name*
- *include your name in the PDF document*
- *include any code you have written as part of the PDF*

Question 1.

In this question we will realise a simple entangling gate, that we will call the $\mathrm{CU_x}$ gate. Instead of the ground-ground (*gg*) Rydberg qubits that we discussed in class, the two qubit states involved are a ground and Rydberg state (*gr* qubit), that can be coupled with Rabi frequency Ω:

$$|\mathrm{g}\rangle = |0\rangle$$
$$|\mathrm{r}\rangle = |1\rangle$$

For this system, single qubit rotations around the x-axis ($\mathrm{U_x}$) have the same form as we discussed in class, but this time involve the levels $|\mathrm{g}\rangle$ and $|\mathrm{r}\rangle$.

The action of this gate applies a "controlled" x-axis rotation to the target qubit dependent on the control qubit state: when the control qubit is in the state $|0\rangle_\mathrm{c}$, the rotation is applied; if the control qubit is in $|1\rangle_\mathrm{c}$, the gate has no action on the target qubit.

Part A (grade: 1-5)

1.1. Write down the 4×4 matrix representation of this gate in the basis

$$\{|0\rangle_c|0\rangle_t,\ |0\rangle_c|1\rangle_t,\ |1\rangle_c|0\rangle_t,\ |1\rangle_c|1\rangle_t\}.$$

1.2. Describe how this gate can be implemented for two adjacent trapped atoms, taking advantage of the Rydberg-Rydberg blockade interaction.

1.3. Sketch the behaviour of the system in response to the applied pulse sequence. You can consider either one of the rubidium ground states (F=1 or F=2) to represent $|g\rangle$ when making the sketch.

1.4. What do we require about the Rydberg excitation shift Δ_{rr} so that this gate functions with high fidelity?

1.5. Describe a potential issue with scalability beyond two qubits for this approach.

Part B (Advanced: grade +0,+1,+2)

1.6. Show that for the right choice of the Rabi frequency and pulse time, this gate can also be used to realise a CNOT gate, if a $R_z(\pi/2)$ is applied to the control qubit prior to applying the $\mathrm{CU_x}$ gate.

1.7. What pulse time τ for the $\mathrm{CU_x}$ is required if the Rabi frequency is 2 kHz?

Question 2.

Consider a pair of trapped ions, that will share collective phonon modes. The ions are prepared initially in the collective motional ground state. The following 3-pulse sequence is applied to the ions to realise an entangling gate between the two ions.

1. Red side band π-pulse on ion A that couples the ground state $|g\rangle$ and the excited electronic state $|e\rangle$.
2. A red sideband 2π-pulse applied to ion B that couples to an $|g\rangle$ to an auxiliary excited state $|a\rangle$.
3. A red sideband π-pulse applied ion A coupling the ground state $|g\rangle$ and the excited electronic state $|e\rangle$.

Part A (grade: 1-5)

1. Determine the action of step 1 of the gate to the basis states of the two-qubit system, $|gg\rangle, |ge\rangle, |eg\rangle, |ee\rangle$.
2. Starting from the result of step 1, determine the action of step 2.
3. Starting from the result of step 2, determine the action of step 3.
4. Write down the resulting 4x4 matrix for this gate and identify its action.
5. Describe how one could realise a CNOT gate using this entangling gate along with single qubit gates (you do not need to calculate the result of the circuit).

Part B (Advanced: grade +0,+1,+2)

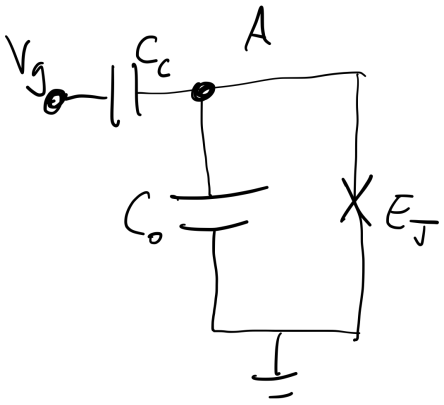
Now consider three qubits labelled A, B, C. Consider how the sequence can be modified to extend the gate to a controlled-controlled gate architecture.

a) Write down the mapping of the qubit states $|ABC\rangle_{initial} \rightarrow |ABC\rangle_{final}$ for the controlled-controlled operation.

b) Determine a modification that realises the operation by adding an additional two pulses to the sequence.

Question 3.

Part A (grade: 1-5)



Consider the circuit for a cooper pair box / transmon shown above. The external node at the left is held at a fixed voltage V_g, which is coupled to the node A by the coupling capacitor C_c . There is a "kinetic" energy associated with the coupling capacitor and voltage, given by $C_c\left(\dot{\Phi} - V_g\right)^2/2$.

3.1. What is the Lagrangian for this circuit, in terms of the flux Φ at A, and $\dot{\Phi}$. This will depend on V_g.

3.2. What is the momentum conjugate to Φ?

3.3. What is the Hamiltonian for the circuit? Write out the Schrodinger equation for this circuit.

3.4. Assuming $V_g = 0$, using a Harmonic oscillator approximation for the cosine potential of the Josephson junction, estimate the ground and first excited state energies.

Part B (Advanced: grade +0,+1,+2)

3.5. Using Mathematica's NDEigensolve function (or other software), numerically compute the lowest 4 eigenenergies for different values of V_g, assuming that $\frac{E_{C_0}}{E_J} = 1$, and plot these.

3.6. What is the smallest value of the energy gap between the ground state and the first excited state, and for what value of V_g does this occur?

Bonus Point:
In 1 page or less, explain how this circuit relates to the Nobel Prize in Physics awarded to Martinis, Clarke, and Devoret, in 2025.

Question 4.

For this problem, consider photonic Bell states entangled in polarisation, i.e. Bell state $|\Psi^-\rangle_{12} = \frac{1}{\sqrt{2}}(|H\rangle_1|V\rangle_2 - |V\rangle_1|H\rangle_2)$ and so on, where the photons are in spatial modes labelled 1,2 (the frequency of the two photons are the same).

Part A (grade: 1-5)

a. The state $|\Psi^-\rangle_{12}$ is input to a Mach-Zender interferometer, which consists of a series of two 50-50 lossless beamsplitters (BS) as shown in Fig. 1 (M_1 and M_2 are mirrors, D_1 and D_2 are single-photon-detectors). That is, one photon of the Bell pair enters one input port of the first beamsplitter at the same time, denoted in Fig. 1 by the black arrows going into BS_1.

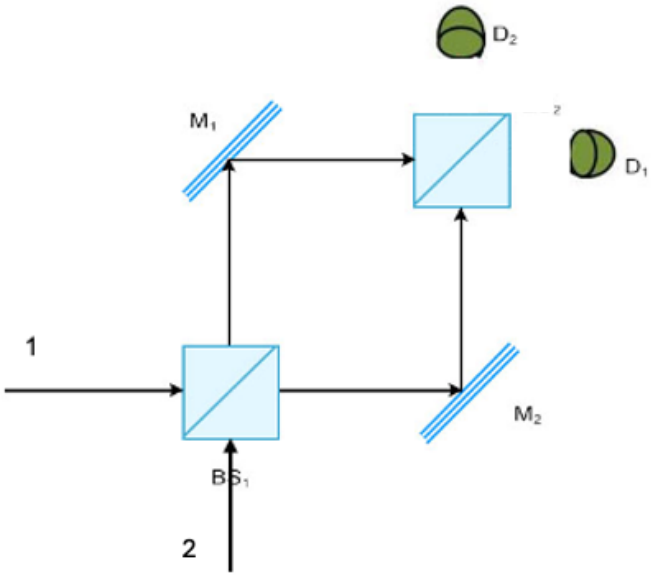


Fig. 1. Mach-Zender interferometer with Bell state input.

What is the output state of this experiment? What is the detection pattern that you observe as a result of this output state? Justify your answer.

b. Show that the other Bell states give rise to the following output states if they are input to a 50:50 beamsplitter (i.e. just one BS, not the setup in Fig. 1):

$$|\Psi^+\rangle_{12} = \frac{1}{\sqrt{2}}(|HV,0\rangle + |0,HV\rangle)$$
$$|\Phi^+\rangle_{12} = \frac{1}{2}(|2H,0\rangle - |0,2H\rangle + |2V,0\rangle - |0,2V\rangle)$$
$$|\Phi^{--}\rangle_{12} = \frac{1}{2}(|2H,0\rangle - |0,2H\rangle - |2V,0\rangle + |0,2V\rangle)$$

Part B (Advanced: grade +0,+1,+2)

Is the detection pattern you observe at the output of the Mach-Zender interferometer experiment in Fig. 1 a result of Hong-Ou-Mandel interference? Why or why not?